THE NOVA RESEARCH CURRICULUM

Answer key · Grade 7 · Term 3

For teachers only. Do not share with students.

Teacher answer key

English

Question bank

#	Outcome	Question	Answer
1	EN-7.4	The first paragraph of a news report should answer ...	only who / **who, what, when, where and why** / the writer's opinion / a joke
2	EN-7.4	News reports put the most important facts ...	at the end / **first** / in the middle / nowhere
3	EN-7.4	A formal complaint should be polite but firm. True or false.	**True**
4	EN-7.4	A summary includes examples and repeated points. True or false.	**False**
5	EN-7.4	Write a headline for a report on your school's sports day.	Model: Blue House Wins Sports Day Trophy
6	EN-7.5	In a debate, a rebuttal is ...	your opening line / **an answer to the other team's point** / the chair's ruling / the vote
7	EN-7.5	The team that supports the motion is the ...	opposition / **proposition** / chair / audience
8	EN-7.5	Write one rebuttal to "Uniforms cost families too much money."	Model: Uniforms often cost less than buying many fashionable clothes, and schools can run uniform-exchange schemes.

Urdu

Question bank

#	Outcome	Question	Answer
1	UR-7.4	کہانی کے آخر میں اکثر کیا ہوتا ہے؟	عنوان / **سبق (اخلاقی نتیجہ)** / تمہید / شعر
2	UR-7.4	مکالمے میں ہر جملے سے پہلے بولنے والے کا نام لکھا جاتا ہے۔ True or false.	**True**
3	UR-7.4	روداد کس زمانے میں لکھی جاتی ہے؟	حال / مستقبل / **ماضی** / کوئی بھی

#	Outcome	Question	Answer
4	UR-7.4	روداد میں کن باتوں کا ذکر ہوتا ہے؟	Model: کب، کہاں، کیا ہوا، کون شریک تھا اور انجام۔
5	UR-7.5	مباحثے میں موضوع کی حمایت کرنے والے کو کیا کہتے ہیں؟	مخالف / **موافق** / صدر / منصف
6	UR-7.5	مباحثے میں وقت کی پابندی ضروری ہے۔ True or false.	**True**

Mathematics

Question bank

#	Outcome	Question	Answer
1	MA-7.6	Circumference of a circle with radius 7 cm (π = 22/7)?	22 cm / **44 cm** / 154 cm / 49 cm
2	MA-7.6	Area of a circle with radius 7 cm (π = 22/7)?	44 cm² / 49 cm² / **154 cm²** / 308 cm²
3	MA-7.6	Volume of a cylinder with r = 7 cm, h = 10 cm (π = 22/7)?	440 cm³ / **1,540 cm³** / 154 cm³ / 4,400 cm³
4	MA-7.6	A prism with cross-section area 12 cm² and length 5 cm has volume …	17 cm³ / **60 cm³** / 120 cm³ / 60 cm²
5	MA-7.6	Doubling the radius of a circle doubles its area. True or false.	**False** (the area becomes four times as large)
6	MA-7.7	Classes 0–10 (f = 2), 10–20 (f = 3), 20–30 (f = 5). Grouped mean?	15 / **18** / 20 / 25
7	MA-7.7	In the same table, the modal class is …	0–10 / 10–20 / **20–30** / none
8	MA-7.7	In a histogram the bars …	have gaps / **touch** / are circles / are always the same height
9	MA-7.7	A bag has 3 red and 5 blue counters. P(red)?	3/5 / **3/8** / 5/8 / 1/3
10	MA-7.7	If P(rain) = 0.3, then P(no rain) = ?	0.3 / **0.7** / 1.3 / 0

Physics

Question bank

#	Outcome	Question	Answer
1	PH-7.4	Weight is measured in …	kilograms / **newtons** / metres / litres
2	PH-7.4	A 50 kg student's weight on Earth (g = 10 N/kg)?	50 N / 5 N / **500 N** / 5,000 N
3	PH-7.4	On the Moon your mass …	becomes six times smaller / **stays the same** / doubles / becomes zero

#	Outcome	Question	Answer
4	PH-7.4	Tides are mainly caused by the ...	wind / Sun's heat / **Moon's gravity** / Earth's core
5	PH-7.4	Seasons happen because ...	the Earth moves closer to the Sun / **the Earth's axis is tilted** / the Moon blocks the Sun / clouds change
6	PH-7.4	Gravity keeps satellites in orbit. True or false.	**True**
7	PH-7.4	Explain the difference between mass and weight.	Model: mass is the amount of matter (kg) and stays the same; weight is the pull of gravity (N) and changes with the planet.

Chemistry

Question bank

#	Outcome	Question	Answer
1	CH-7.3	In salt water, salt is the ...	solvent / **solute** / solution / precipitate
2	CH-7.3	Which is insoluble in water?	sugar / salt / **sand** / coffee powder
3	CH-7.3	A solution that can dissolve no more solute is ...	dilute / **saturated** / insoluble / empty
4	CH-7.3	For most solids, solubility increases as temperature ...	falls / **rises** / stays the same / disappears
5	CH-7.3	Fizzy drinks go flat faster when warm because gases dissolve less in warm water. True or false.	**True**
6	CH-7.3	In a fair test, how many variables do you change?	**one** / two / three / all of them
7	CH-7.3	Name two variables to keep the same in the sugar test.	Model: the amount of water and the amount of stirring (also: type of sugar, size of spoon).

Biology

Question bank

#	Outcome	Question	Answer
1	BI-7.3	Dengue is spread by ...	dirty water / **mosquitoes** / touch / food
2	BI-7.3	Polio is caused by a ...	bacterium / fungus / **virus** / parasite
3	BI-7.3	Which cells make antibodies?	red blood cells / platelets / **white blood cells** / nerve cells
4	BI-7.3	Antibiotics work against viruses. True or false.	**False** (they work against bacteria)

#	Outcome	Question	Answer
5	BI-7.3	A vaccine contains ...	a strong live germ / **a harmless form or part of a germ** / antibiotics / vitamins only
6	BI-7.3	Skin is one of the body's first defences. True or false.	**True**
7	BI-7.3	Give two ways to stop dengue spreading.	Model: remove standing water and use mosquito nets or repellent.
8	BI-7.3	Why are you less likely to catch measles twice?	Model: memory cells remember the virus and the body makes antibodies quickly.

History, Geography and Pakistan Studies

Question bank

#	Outcome	Question	Answer
1	HI-7.2	Which Europeans arrived first by sea?	English / Dutch / **Portuguese** / French
2	HI-7.2	The English East India Company was given its charter in ...	1498 / **1600** / 1757 / 1857
3	HI-7.2	The Battle of Plassey was fought in ...	1707 / **1757** / 1764 / 1799
4	HI-7.2	Mir Jafar's betrayal helped the British win at Plassey. True or false.	**True**
5	HI-7.2	Tipu Sultan died fighting the British in ...	1757 / 1764 / **1799** / 1857
6	GE-7.2	Sui is famous for ...	salt / coal / **natural gas** / copper
7	GE-7.2	Khewra is famous for ...	**salt** / gold / oil / marble
8	GE-7.2	Which is a renewable energy source?	coal / gas / oil / **wind**
9	GE-7.2	Name two ways Pakistan can use its energy more wisely.	Model: more solar and wind power, and saving electricity at home (also: fixing gas leaks, efficient appliances).

Islamiyat and Ethics

Question bank

#	Outcome	Question	Answer
1	IS-7.3	Which is haram to eat?	fish / vegetables / **pork** / milk
2	IS-7.3	Riba means ...	charity / **interest** / trade / zakat

#	Outcome	Question	Answer
3	IS-7.3	Cheating in weighing is forbidden in Islam. True or false.	**True**
4	IS-7.3	Give two examples of haram earning.	Model: bribery and cheating customers (also: theft, interest).
5	IS-7.4	Who began the Hijri calendar?	Hazrat Abu Bakr (RA) / **Hazrat Umar (RA)** / Hazrat Uthman (RA) / Hazrat Ali (RA)
6	IS-7.4	Hazrat Uthman (RA) is called Ghani because he was very ...	strong / **generous** / old / quiet
7	IS-7.4	The fourth caliph was ...	Hazrat Umar (RA) / Hazrat Uthman (RA) / **Hazrat Ali (RA)** / Hazrat Abu Bakr (RA)
8	IS-7.4	Hazrat Aisha (RA) narrated many ahadith. True or false.	**True**
9	ET-7.2	Name two ways to protect the environment at school.	Model: recycle paper and turn off taps and lights (also: plant trees, use less plastic).

Computer Science and skills

Question bank

#	Outcome	Question	Answer
1	CS-7.3	The examples an AI learns from are called ...	software / **training data** / hardware / output
2	CS-7.3	If training data leaves out some groups, the AI may be ...	faster / **biased** / smaller / offline
3	CS-7.3	An AI answer that sounds confident is always correct. True or false.	**False**
4	CS-7.4	A device that sends data between networks is a ...	monitor / **router** / printer / keyboard
5	CS-7.4	Data on the internet travels in small pieces called ...	bricks / **packets** / files only / signals only
6	CS-7.4	You should share a one-time code (OTP) with someone who calls from "the bank". True or false.	**False**
7	CS-7.4	Give two signs of a phishing message.	Model: it asks for a password or code, and it creates panic or urgency (also: odd links, spelling mistakes, unknown sender).
8	FL-7.1	Islamic banking avoids ...	saving / **interest (riba)** / trade / profit